

BLITTER DISK II ADAPTER

Original Apple Disk II to USB Filesystem Bridge

Revision A - Prototype Schematic Package

Project conceived and directed by Darren Crawford

For live mounting of real Apple II DOS 3.3 and ProDOS disks on modern Windows and Linux PCs

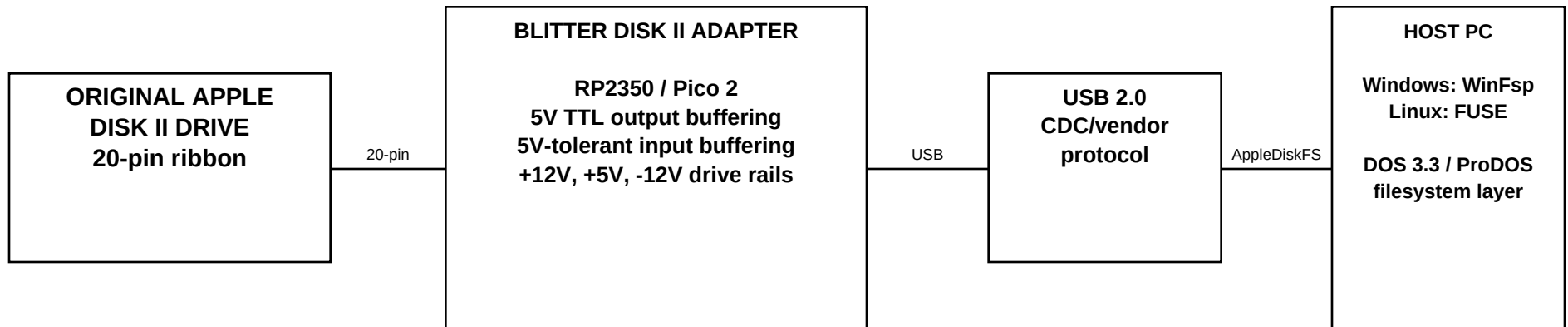
ENGINEERING PROTOTYPE - NOT YET BENCH-QUALIFIED ON VINTAGE HARDWARE

Verify all rails and logic states with the Disk II disconnected before attaching an original drive.

Never hot-plug the 20-pin Disk II cable.

1. System Architecture

The PC sees a live filesystem; the adapter handles vintage drive timing and power.



Design goal: insert a real Apple II floppy, mount it in Windows or Linux, and use ordinary file operations. The RP2350 performs deterministic motor, head-step, read-data, and write-data timing. The host parses DOS 3.3 or ProDOS and exposes files to Explorer or a Linux mount point.

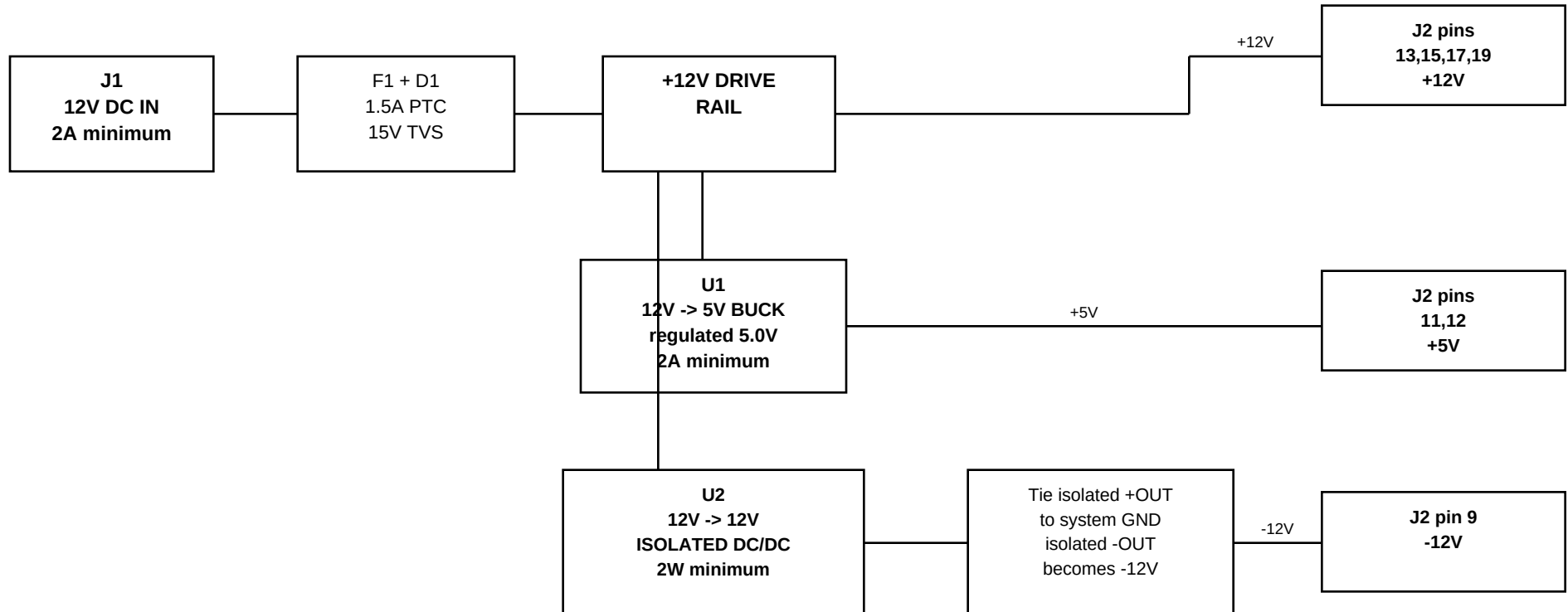
Power architecture

USB powers only the Pico 2 and logic. The vintage Disk II is powered from a separate regulated 12V adapter. The board passes protected +12V and derives +5V and -12V locally for the drive.

2. Power Schematic

External 12V supply for the drive; USB and drive power remain functionally separated.

Recommended Rev A approach: use a quality regulated 12V/2A-or-greater wall adapter. Use a known-good 5V buck converter adjusted to exactly 5.0V before connection. Use an isolated 12V-to-12V converter for the negative rail; referencing its positive output to ground creates -12V.

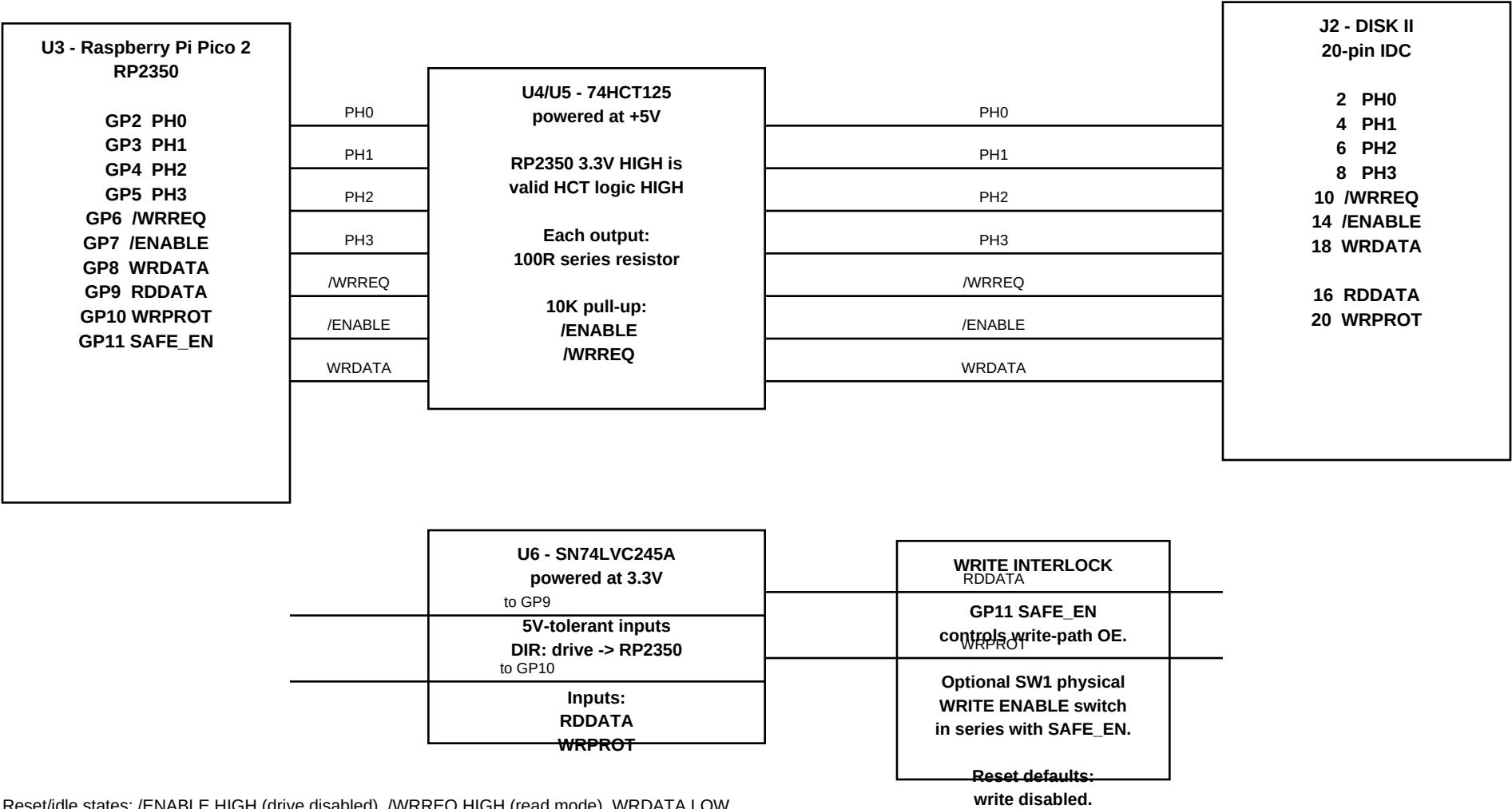


Common ground: J1 negative, Pico GND, logic GND and Disk II pins 1/3/5/7.

Local decoupling at J2: +12V 100uF + 100nF; +5V 100uF + 1uF + 100nF; -12V 10uF + 100nF. Add 100nF at each logic IC.

3. Logic Interface

3.3V RP2350 logic is translated to TTL-compatible 5V drive signals.



Reset/idle states: /ENABLE HIGH (drive disabled), /WRREQ HIGH (read mode), WRDATA LOW.
Firmware must not assert write signals until the host explicitly arms writing.

4. Disk II 20-pin Connector

J2: keyed/shrouded 2x10, 0.1-inch. Mark pin 1 permanently.

Pin	Signal	Purpose
1	GND	Ground
2	PH0	Stepper phase 0
3	GND	Ground
4	PH1	Stepper phase 1
5	GND	Ground
6	PH2	Stepper phase 2
7	GND	Ground
8	PH3	Stepper phase 3
9	-12V	Drive negative supply
10	/WRREQ	Write request, active low
11	+5V	Drive +5V
12	+5V	Drive +5V
13	+12V	Drive +12V
14	/ENABLE	Drive enable, active low
15	+12V	Drive +12V
16	RDDATA	Read data from drive
17	+12V	Drive +12V
18	WRDATA	Write data to drive
19	+12V	Drive +12V
20	WRPROT	Write-protect sense from drive

BOARD HEADER / MATING FACE									
1	3	5	7	9	11	13	15	17	19
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
2	4	6	8	10	12	14	16	18	20
PIN 1 = red-stripe side of ribbon cable									

A reversed cable can place drive power on logic pins. Prefer a shrouded/keyed connector and add an obvious pin-1 triangle on the PCB silkscreen.

Board -> Drive: PH0, PH1, PH2, PH3, /WRREQ, /ENABLE, WRDATA.
Drive -> Board: RDDATA, WRPROT.

5. Bill of Materials

Prototype-friendly Rev A parts list.

Ref	Part	Specification / notes	Qty
U3	Raspberry Pi Pico 2	RP2350 module	1
U4,U5	74HCT125	Quad 5V HCT buffer	2
U6	SN74LVC245A	5V-tolerant input buffer, 3.3V	1
U1	5V buck converter	12V in, 5.0V out, >=2A	1
U2	Isolated DC/DC	12V in / 12V out, >=2W	1
J1	DC jack	5.5x2.1mm center-positive	1
J2	2x10 IDC header	0.1in, keyed/shrouded	1
F1	Resettable PTC	~1.5A hold	1
D1	TVS diode	~15V standoff	1
R1-R7	100 ohm	Series resistors on outputs	7
R8,R9	10K	Pull-ups on /ENABLE, /WRREQ	2
R10	10K	SAFE_EN pull-down	1
C1,C2	100uF	Bulk decoupling	2
C3+	100nF	Logic/rail decoupling	6+
SW1	SPST switch	Physical write-enable interlock	1

Use a PCB or well-built perfboard with short signal runs and a solid ground. Do not use a solderless breadboard for the Disk II power rails.

6. Bring-up Checklist

Perform these checks before connecting an original Apple Disk II.

1. Leave J2 empty. Connect USB only. Confirm the Pico boots and write remains disabled.
2. Apply the external 12V supply.
3. Measure J2 pins 13/15/17/19: approximately +12V relative to GND.
4. Measure J2 pins 11/12: +5.0V relative to GND.
5. Measure J2 pin 9: approximately -12V relative to GND.
6. Verify J2 pins 1/3/5/7 are ground.
7. Verify /ENABLE and /WRREQ idle HIGH and WRDATA idles LOW.
8. Exercise PH0-PH3 using test firmware and confirm valid 0V/5V switching at J2.
9. Confirm SAFE_EN cannot enable writing unless the optional physical WRITE ENABLE switch is ON.
10. Power everything off before attaching the Disk II cable. Confirm pin-1 orientation twice.

Host software target

Windows: WinFsp-backed volume, for example A:\ or B:\. Linux: FUSE-backed mount such as /media/blitterdisk. The host should auto-detect DOS 3.3 versus ProDOS, cache sector/block access, preserve Apple file type and auxiliary type metadata, and flush changes safely before eject.

Publication credit

Project name: Blitter Disk II Adapter. Project conceived and directed by Darren Crawford. Community review and independent electrical verification are strongly encouraged before production hardware is made.